

Stage 8 — Are the real-artifact labels right, and does the law hold for a strong model?

Where the Shortcut Sits: progress report 8

In one sentence. Checked without a clinician — against three public expert annotation sets and two independent automatic labellers — the dermoscopy labels of Stage 3 put hair on the correct side of the lesion boundary but mark hair on many images that experts call hair-free, and removing every image an independent source contradicts *increases* the crossover ($+0.147$ [$+0.111, +0.185$] $\rightarrow +0.199$ [$+0.164, +0.233$]); the thyroid caliper labels agree with two independent labellers wherever those agree with each other, but the registered cleaning rule removed too many images for its test to run; the ovary and capsule labels could not yet be checked; and a fine-tuned thyroid model 0.004 AUROC below the published benchmark shows every registered harm of masking, including a sensitivity loss from 0.744 to 0.223 in the 78 malignant nodules with a caliper on the nodule, a subgroup now registered before any result; about 30% of that loss survives even when each model’s threshold is re-tuned on the test set itself.

Notation

$Y \in \{0, 1\}$	diagnosis: 1 = melanoma, malignant nodule, tumour with a solid component, erosion.
$A \in \{0, 1\}$	artifact present: a real artifact: hair, sonographer caliper, capsule debris.
ROI	region of interest: lesion, nodule, tumour, union of expert lesion boxes; drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (artifact in the ROI); Trap B: $r < 0.1$ (outside).
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real artifacts: half of each class) or absent (synthetic artifacts).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{Trap B}} - [\text{mask} - \text{ERM}]_{\text{Trap A}}$ of reversed-test AUROC (Stage 3).
CI	95% crossed seed $\times$ image bootstrap interval (Stage 1), 10,000 replicates, for every model result; label agreement: 95% bootstrap interval over images, 2,000 replicates.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted into the same image.
Cell	our label of an image: artifact-free, Trap A, Trap B, or neither (overlap between the two thresholds).
Contradicted	an informative independent source places the image in another cell (artifact present vs absent, or the other side of the ROI boundary).
e_A, e_B	share of Trap-A (Trap-B) images that a source places on the other side of the boundary.
κ	Cohen's kappa.
Hard pairs	malignant nodules with an in-ROI caliper vs benign nodules without; they conflict with the training association (Stage 5).
OP1–OP5	decision thresholds fixed on validation data (Stage 5).

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1 Where this stage starts

Stages 1–7 established the location law — ROI masking removes an out-of-ROI shortcut and keeps, or amplifies, an in-ROI one — and showed that it holds across masking implementations, is graded in the real overlap and appears on an external test set (Stage 7). Two objections remained that the earlier stages could not answer with their own data:

1. *“Nobody checked the real-artifact labels.”* The real traps of Stage 3 rest on automatic labels: published hair masks (Wegley et al., 2026), a rule-based caliper detector (thyroid, ovary), a contamination probe (capsule) and lesion masks that are partly U-Net predictions. The audit package of Stage 6 was prepared for a clinician, and no clinician is available to rate it.
2. *“The thyroid model is weaker than published ones, and the subgroup was chosen after the fact.”* The frozen probe of Stage 5 reaches a test AUROC of 0.735 on the official patient-disjoint split, below the fine-tuned models of Gong et al. (2022) (0.773–0.799); the 78 malignant nodules with an in-ROI caliper, in which masking cost most sensitivity, were defined post hoc.

This stage answers both with one registration (Part A: labels; Part B: a fine-tuned model), committed with three amendments before any analysis.

2 Questions

- Q8a** How often do independent sources contradict our artifact labels, and do the errors differ by diagnosis?
- Q8b** Does the crossover survive on the images that no independent source contradicts?
- Q8c** Does a fine-tuned model close to the published benchmark show the harms that masking caused the frozen probe?
- Q8d** Does the subgroup harm hold when the subgroup is registered in advance?
- Q8e** Is the sensitivity loss an artefact of thresholds transported from validation to test?

Design decisions *Why a clinician-free check is enough for these claims.* Whether a hair crosses a lesion, a caliper lies on a nodule or debris lies inside a box is a visual fact, not a diagnosis, and the claims of Stage 3 need only these facts. Moreover, if trap membership is misclassified independently of the diagnosis, the observed crossover is attenuated, $C_{\text{obs}} \approx (1 - e_A - e_B) C_{\text{true}}$: random label errors shrink a crossover and cannot create one. The checks therefore measure the error rates, test whether they differ by diagnosis, and re-run the traps without the contradicted images. *Sources.* Public expert annotations where they exist: DermArtifactDB for hair presence (der, 2026), IMA++ lesion masks (ima, 2024), the hair masks of Kabir et al. and expert contamination masks for capsule frames (cap). No public caliper annotation exists, so two independent automatic labellers are used for calipers: BUSClean, an ultrasound annotation detector (bus), and MedGemma, a medical vision–language model (Google), both run locally. *Safeguards* (Amendments 1–2): images on which one of our own labellers was trained are excluded from every comparison; a source is analysed only if it covers at least 200 of our images; a model labeller counts only if its κ is at least 0.40 on the images on which the two other caliper labels agree; BUSClean is declared invalid if it fires on more than half of our caliper-free images; neither model labeller is ever shown our artifact masks. *Rating.* A blinded rating of the audit sample by the author (a non-clinician) is registered (A4); it has not been done yet and does not enter the results below. *Part B.* The recipe is chosen on validation images only; the benchmark is the lowest published cross-entropy baseline on the same split (0.773).

3 Methods

A0 — sources and coverage Each source’s licence was read from the source itself and recorded before use. Matching: ISIC by image identifier; capsule frames by perceptual hash (Hamming distance ≤ 2 , as in Stage 1). IMA++ is used as a majority mask over its annotators; its masks and anything derived from them at pixel level stay on the GPU machine (CC BY-NC-ND). The coverage report was committed before any agreement was computed.

A1 — agreement Hair: κ of our hair-free status (≤ 30 hair pixels) with DermArtifactDB; Dice of our hair masks with the Kabir masks. Lesion masks: Dice with the IMA++ majority mask, and the trap class recomputed with it (like for like at 518 px). Calipers: κ of presence with BUSClean and with MedGemma; for location, agreement of our class with BUSClean’s detected positions and with MedGemma’s answer to “is any caliper mark inside the green contour?” on the image with only the nodule outline drawn. MedGemma: fixed prompts, greedy decoding, answer parsed as yes/no. Every agreement is reported for $Y = 0$ and $Y = 1$ separately.

A2 — traps on uncontradicted labels Contradicted images are removed; images without an independent source are kept. The Stage 3 protocol is re-run unchanged (DINOv2 ViT-B/14 at 518 px, same matching, seeds and folds, ERM and mask arms, crossed bootstrap) with the count gate of Stage 7 (≥ 25 images per matched cell). HA2: crossover > 0 in every cohort that passes the gate (Holm).

A3 — differential error Error rates of our cells against every informative source by trap and diagnosis; differential error is flagged if the rates for $Y = 1$ and $Y = 0$ differ by more than 0.10 with a CI excluding zero, and a flagged cohort’s conclusion then rests on A2. The attenuation correction is reported only for unflagged cohorts.

Part B — fine-tuned thyroid model Twelve recipes (ConvNeXt-T, ResNet-50, DINOv2 ViT-B/14 $\times$ two learning rates $\times$ 8 or 20 epochs; 224 px, the training procedure of Stage 5) are compared by validation AUROC of ERM, seed 42. The chosen recipe is then trained on the natural split (5 seeds $\times$ ERM, mask, balanced, mask + balanced; validation predictions saved for the operating points) and in the thyroid traps (2 environment seeds $\times$ 5 folds = 10 bootstrap clusters). FT0 (descriptive): mean ERM test AUROC ≥ 0.773 . FT1: hard-pair AUROC, mask – ERM < 0 . FT2: sensitivity CI below zero at ≥ 3 of OP2–OP5. FT3: at OP2, sensitivity of the malignant nodules with an in-ROI caliper, mask – ERM < 0 . FT4: trap crossover > 0 . Holm over FT1–FT4.

Round 7 — thresholds matched on the test set Registered after Part B was known and before this analysis ran (`docs/PREREGISTRATION_ROUND7.md`); saved predictions only. For each seed and arm the threshold is chosen on the test images themselves — the best case for any re-tuning at the deployment site — so that ERM and the masked model reach the same test specificity (0.80: registered; 0.90 and matched sensitivity 0.80: descriptive). TM1 (fine-tuned model) and TM2 (frozen probe): sensitivity of the 78 conflicting malignant nodules, mask – ERM < 0 (Holm over two); thresholds are re-estimated in every bootstrap replicate.

E1–E2 — exploratory Decided after A1–A3 were known and reported separately: a consensus view in which an image counts as contradicted only if both caliper labellers place it in the same cell and that cell differs from ours (E1), and FT3 restricted to the subgroup nodules that both labellers confirm (E2).

4 Experiments

Table 1: Experiments of this stage. `docs/PREREGISTRATION.ROUND6.md` and its three amendments were committed before any analysis of this round; the coverage report (`4c3263d`, 2026-09-28 16:29), the fine-tune recipe (`b13bcd`, 2026-09-28 16:29) and the hash of the A4b key (`4b32a1f`, 2026-09-28 16:34) were committed before the steps that use them. Commit times are set by the committing machine and are not independent proof of order.

Experiment	Status	First commit (local time)	Support criterion
A0 sources, licences, coverage	registered (round 6); before results (round 6); coverage committed before A1	607900c, 14:27, 2026-09-28	≥ 200 matched images: analysed; 50–199: descriptive; < 50 : not feasible
A1 agreement with independent sources	registered (round 6); before results (round 6)	607900c, 14:27, 2026-09-28	descriptive; a model labeller is used only if $\kappa \geq 0.40$ (Amendment 2: validity guard)
A2 traps on uncontradicted labels (HA2)	registered (round 6); before results (round 6); primary of Part A	607900c, 14:27, 2026-09-28	crossover > 0 in every cohort that passes the count gate (Holm)
A3 differential error, attenuation	registered (round 6); before results (round 6)	607900c, 14:27, 2026-09-28	flag: $ e_{Y=1} - e_{Y=0} > 0.10$, CI excluding 0; correction only for unflagged cohorts
A4 blinded rating by the author (A4b optional)	registered (round 6); before results (round 6); not yet done	607900c, 14:27, 2026-09-28	intra-rater κ ; error rates of our cells; judges a single model labeller (Amendment 2)
B1 fine-tune recipe (validation only)	registered (round 6); before results (round 6); choice committed before B2	607900c, 14:27, 2026-09-28	highest validation AUROC of 12 recipes
B2–B3 natural split and traps (FT0–FT4)	registered (round 6); before results (round 6)	607900c, 14:27, 2026-09-28	FT0 descriptive (≥ 0.773); FT1–FT4 one-sided, Holm over four
Round 7: thresholds matched on the test set (TM1–TM2)	registered (round 7); before results (round 7)	cbf4229, 20:37, 2026-09-28	conflicting sensitivity at test specificity 0.80, mask – ERM < 0 (Holm over two)
E1–E2 consensus labels	post hoc (exploratory; decided after A1–A3 were known)	—	descriptive; not a test

5 Results

5.1 Coverage

Table 2. Five sources pass the coverage gate. IMA++ covers 241 of our dermoscopy images once the 2,503 ISIC 2018 training images of our lesion U-Net are excluded. The capsule masks match 51 of our frames, of which 19 trained our contamination probe and 3 already carry an expert mask in Stage 3, leaving 29: the capsule labels cannot be checked.

Table 2: Independent label sources and their coverage of our cohorts after the exclusions of Amendment 1 (training images of our own labellers). No image of the thyroid or ovary cohorts left the GPU machine: BUSClean and MedGemma ran locally.

Cohort	Source	Used for	Licence	Matched images	Gate
Dermoscopy hair	DermArtifactDB	hair present / absent	CC BY 4.0	20,519	analysed
Dermoscopy hair	IMA++	lesion masks, 1–5 annotators	cc-by-nc-nd-4.0	241	analysed
Dermoscopy hair	Kabir hair masks	hair masks	CC BY 4.0	485	analysed
Capsule debris	Multicentre clear/contaminated capsule masks	contamination masks	CC BY 4.0	29	not feasible
Thyroid calipers	BUSClean	annotation detector	MIT	3,493	analysed
Ovary calipers	BUSClean	annotation detector	MIT	1,202	analysed
Thyroid calipers	MedGemma 1.5 4B-it	vision–language answers (presence, location)	model terms of use	3,492	analysed
Ovary calipers	MedGemma 1.5 4B-it	vision–language answers (presence, location)	model terms of use	1,198	analysed

5.2 Dermoscopy hair: position right, presence over-called

Table 3. *Presence.* Agreement with DermArtifactDB is low, $\kappa = 0.076$ [0.070, 0.082], and the disagreement is one-sided. Our artifact-free cell is clean: 97% of its 627 images are hair-free in DermArtifactDB too. But our masks exceed the registered threshold of 30 hair pixels on 93% of the 8,864 images DermArtifactDB calls hair-free (hair on 97% of images by our threshold, on 57% by DermArtifactDB). Inside the traps DermArtifactDB reports no hair for 53% of Trap A and 51% of Trap B images, more often for melanomas (Trap A: 63% against 50%; Trap B: 65% against 48%). The difference is flagged as differential error in both traps (+0.133 [+0.100, +0.168] and +0.166 [+0.134, +0.198]), so by the registered rule the dermoscopy conclusion rests on A2. *Position.* Where hair is present, it is placed correctly. Our lesion masks agree with the IMA++ majority masks (Dice 0.851 [0.830, 0.870]), the trap class recomputed with IMA++ agrees for 0.841 [0.791, 0.887] of images, and IMA++ puts the hair on the other side of the boundary for 0/15 Trap-A and 4/158 Trap-B images. Our hair masks overlap the independent Kabir masks with Dice 0.618 [0.600, 0.637] — thin structures, so pixel agreement is modest — give the same trap cell for 87% of trap images and the opposite side for 0/31 Trap-A and 1/233 Trap-B images. The measured location error $e_A + e_B$ is 0.025 (IMA++) and 0.004 (Kabir).

Table 3: Dermoscopy: agreement of our labels with three independent sources (95% bootstrap CIs over images, 2,000 replicates). The Kabir images are all hair images and our masks mark hair on every one of them, so κ is not informative there; the raw agreement is given instead.

Source	Quantity	Images	All	$Y = 0$ (benign)	$Y = 1$ (melanoma)
DermArtifactDB	hair present (ours: > 30 hair pixels)	20,519	κ 0.076 [0.070, 0.082]	κ 0.083 [0.076, 0.090]	κ 0.051 [0.041, 0.060]
IMA++ (majority mask)	lesion mask (Dice)	241	0.851 [0.830, 0.870]	0.851 [0.825, 0.872]	0.853 [0.819, 0.888]
IMA++	trap class (r recomputed)	239	0.841 [0.791, 0.887]	0.855 [0.807, 0.899]	0.750 [0.594, 0.875]
Kabir et al.	hair mask (Dice)	485	0.618 [0.600, 0.637]	0.617 [0.595, 0.638]	0.629 [0.592, 0.665]
Kabir et al.	hair present	485	agreement 97.3%		

Traps on uncontradicted labels Removing the contradicted images (42% of Trap A, 43% of Trap B) raises the crossover from +0.147 [+0.111, +0.185] to +0.199 [+0.164, +0.233] (Table 4, Figure 1; HA2 supported). This is what attenuation predicts: images whose “hair” is not hair dilute the effect, and removing them sharpens it.

Table 4: Traps re-run on the images no informative source contradicts (A2; DINOv2 ViT-B/14, Stage 3 protocol, crossed 95% CIs, Holm over the cohorts that ran). For capsule and ovary no image was removed, so the re-run reproduces Stage 3 and adds no evidence about the labels.

Cohort	Stage 3 crossover	Uncontradicted labels	Removed (Trap A / Trap B)	HA2
Dermoscopy hair (ISIC 2019)	+0.147 [+0.111, +0.185]	+0.199 [+0.164, +0.233]	42% / 43%	supported
Thyroid calipers	+0.239 [+0.196, +0.280]	—	61% / 87%	count gate not met (Trap B)
Capsule debris	+0.377 [+0.327, +0.426]	+0.377 [+0.327, +0.425]	0% / 0%	supported; nothing removed (no source analysable)
Ovary calipers	+0.170 [+0.087, +0.252]	+0.170 [+0.089, +0.255]	0% / 0%	supported; nothing removed (no informative source yet)

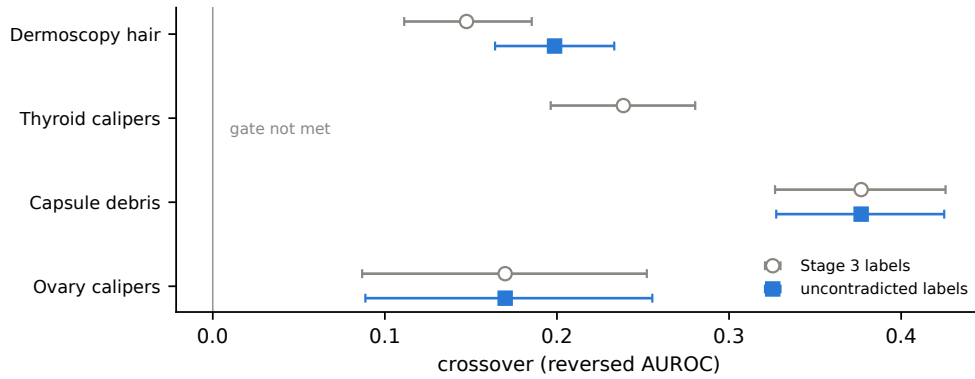


Figure 1: Crossover on the Stage 3 labels (open circles) and on the images no informative source contradicts (squares), crossed 95% CIs. Capsule and ovary: nothing was removed; thyroid: the registered cleaning left too few Trap-B images.

5.3 Thyroid calipers: two informative labellers with opposite biases

BUSClean passes the validity guard (it fires on 5.2% of our caliper-free images), and both labellers pass the informativeness rule ($\kappa = 0.505$ and 0.654). Their presence agreement with our detector is $\kappa = 0.483$ [0.456, 0.509] (BUSClean) and 0.663 [0.638, 0.686] (MedGemma); location agreement is 0.818 [0.790, 0.845] on the 770 images where BUSClean returns positions and 0.653 [0.627, 0.677] on 1,419 images for MedGemma. Table 5 shows why the registered cleaning failed: the two labellers err in opposite directions. MedGemma reports a caliper on 99% of our Trap-A images but also on 29% of our caliper-free images; BUSClean rarely fires on caliper-free images but finds an annotation on only 56% of Trap-A images. Under the registered rule — remove an image if *any* informative source disagrees — the two biases add up: 61% of Trap A and 87% of Trap B are removed, the cleaned Trap B keeps 29 benign and 14 malignant caliper images, below the gate of 25, and HA2 cannot be tested for thyroid.

Table 5: Thyroid: how often each independent labeller reports a caliper in each of our cells, and the share of images the registered rule removes (an image is removed if *any* informative labeller places it in another cell).

Our cell	Images	BUSClean: annotation found	MedGemma: caliper “yes”	Contradicted (registered rule)
caliper-free	1,801	5%	29%	30%
Trap A (in the nodule)	1,150	56%	99%	61%
Trap B (outside)	326	37%	87%	87%

Differential error The measured location errors are $e_A = 0.106$, $e_B = 0.055$ against BUSClean and $e_A = 0.352$, $e_B = 0.270$ against MedGemma. BUSClean’s disagreement in Trap A is smaller for malignant nodules (-0.163 $[-0.229, -0.099]$), which is flagged; MedGemma’s is not (-0.042 $[-0.108, +0.018]$ and $+0.042$ $[-0.074, +0.156]$). The cohort is therefore flagged and no attenuation correction is reported.

Where the two labellers agree (E1, exploratory) Table 6. BUSClean and MedGemma give the same cell for 1,865 of 3,277 images. On these, they agree with our cell for 94%: caliper-free 97%, Trap A 94%, Trap B 48%. Counting an image as contradicted only when both labellers agree against us removes 2% of caliper-free images, 2% of Trap A and 14% of Trap B (11% of benign, 21% of malignant). Presence and in-nodule calipers are well supported; the weakest cell is Trap B, calipers outside the nodule. Re-running the traps without the images that both labellers contradict (E1) keeps 202 benign and 78 malignant Trap-B caliper images, passes the gate, and gives a crossover of $+0.257$ $[+0.209, +0.304]$, against $+0.239$ $[+0.196, +0.280]$ on the Stage 3 labels.

Table 6: Thyroid, **exploratory**: the images on which BUSClean and MedGemma give the same cell. An image counts as contradicted only if both place it in the same cell and that cell differs from ours.

Our cell	Both agree with other	labellers with each	... and our cell	with	Contradicted by both	benign	malignant
caliper-free	1,301		97%		2%	2%	2%
Trap A (in the nodule)	475		94%		2%	3%	1%
Trap B (outside)	89		48%		14%	11%	21%

The registered subgroup Of the 78 malignant test nodules that our detector gives an in-ROI caliper (FT3), no nodule is contradicted by both labellers, both confirm the caliper inside the nodule for 36, MedGemma places it inside for 56 and never reports no caliper, and BUSClean finds no annotation on 19. Restricted to the 36 doubly confirmed nodules (E2), the OP2 sensitivity loss is -0.394 $[-0.570, -0.236]$ for the fine-tuned model and -0.189 $[-0.364, -0.011]$ for the frozen probe (the other 42: -0.629 $[-0.761, -0.436]$ and -0.529 $[-0.676, -0.360]$).

5.4 Ovary calipers: one usable labeller

BUSClean fails the validity guard: it fires on 90.5% of our caliper-free ovary images, so on these screens it detects on-screen annotation and text, not calipers, and is not used. MedGemma is the only labeller left; its presence agreement with our detector is low, $\kappa = 0.194$ $[0.148, 0.247]$ (benign 0.294 $[0.219, 0.375]$, malignant 0.095 $[0.042, 0.153]$), and at least one of the two is often wrong on these images. Which one is wrong is exactly what the informativeness rule decides, and with a single model labeller the registered arbiter is the author’s blinded rating (Amendment 2), which has not been done. No ovary image is removed, and the A2 re-run reproduces Stage 3 (Table 4); it is not evidence about the labels.

5.5 Capsule debris: not checkable

With 29 matched frames the capsule comparison is not feasible by the coverage gate. All 29 expert masks of these frames mark no contaminated pixel, including the 4 frames our probe places in a trap; with so few frames this is not interpreted. As for ovary, the A2 re-run reproduces Stage 3.

5.6 Part B: a fine-tuned thyroid model near the benchmark

Recipe Table 7. ConvNeXt-T (ImageNet-22k), learning rate 10^{-4} , 8 epochs had the highest validation AUROC (0.865; best DINOv2 ViT-B/14 recipe 0.850) and was committed before the final models were trained.

Table 7: Recipe selection on validation images only (B1); no test image was scored. The chosen recipe (bold) was committed before any final model was trained.

Architecture	Learning rate	Epochs	Validation AUROC (ERM, seed 42)
ConvNeXt-T (IN-22k)	3e-5	8	0.849
ConvNeXt-T (IN-22k)	3e-5	20	0.842
ConvNeXt-T (IN-22k)	1e-4	8	0.865
ConvNeXt-T (IN-22k)	1e-4	20	0.865
ResNet-50	3e-5	8	0.658
ResNet-50	3e-5	20	0.723
ResNet-50	1e-4	8	0.723
ResNet-50	1e-4	20	0.781
DINOv2 ViT-B/14	3e-5	8	0.838
DINOv2 ViT-B/14	3e-5	20	0.850
DINOv2 ViT-B/14	1e-4	8	0.715
DINOv2 ViT-B/14	1e-4	20	0.804

Hypotheses Table 8. The mean ERM test AUROC is 0.769, 0.004 below the benchmark of 0.773: by the registered rule the model is *below benchmark*; the frozen probe reached 0.735. All four tests are supported after Holm adjustment. On shortcut-conflicting pairs masking lowers the AUROC by -0.102 $[-0.178, -0.026]$ (FT1), while on all pairs the change is -0.024 $[-0.064, +0.016]$ and on aligned pairs $+0.017$ $[-0.032, +0.065]$. Group balancing recovers the conflicting pairs (balanced – mask $+0.088$ $[+0.019, +0.158]$), masking and balancing together do not (-0.016 $[-0.058, +0.019]$). The trap crossover is $+0.187$ $[+0.111, +0.268]$ (FT4).

Table 8: Part B: the chosen fine-tuned model on the official patient-disjoint thyroid split (5 seeds $\times$ 4 arms) and in the thyroid traps (10 bootstrap clusters). Crossed bootstrap; Holm over FT1–FT4.

	Hypothesis	Estimate [95% CI]	p (one-sided)	Holm p	Verdict
FT0	ERM test AUROC ≥ 0.773 (descriptive)	0.769	—	—	below benchmark
FT1	hard-pair AUROC, mask – ERM < 0	-0.102 $[-0.178, -0.026]$	0.0040	0.0040	supported
FT2	sensitivity loss (CI < 0) at ≥ 3 of OP2–OP5	4 of 4	0.0005	0.0015	supported
FT3	OP2 sensitivity, malignant with in-ROI caliper, mask – ERM < 0	-0.521 $[-0.642, -0.382]$	0.0005	0.0015	supported
FT4	thyroid trap crossover > 0	$+0.187$ $[+0.111, +0.268]$	0.0001	0.0004	supported

Operating points Table 9 and Figure 2. Masking lowers sensitivity at every operating point, at OP2 from 0.747 to 0.371 (-0.376 $[-0.462, -0.297]$), and raises specificity (from 0.672 to 0.897): thresholds fixed on validation images land at a more conservative point on the test set once the model is trained on masked images. In the registered subgroup of 78 malignant nodules with a caliper on the nodule, sensitivity at OP2 falls from 0.744 to 0.223 (-0.521 $[-0.642, -0.382]$; FT3). This test was registered before any result of this model existed.

Table 9: Fine-tuned model, official patient-disjoint thyroid test split: masking at five validation-fixed operating points (crossed bootstrap, thresholds re-estimated in every replicate).

Operating point (fixed on validation)	Sensitivity ERM $\rightarrow$ mask	Δ sensitivity	Δ specificity	Sensitivity, malignant with in-ROI caliper (ERM $\rightarrow$ mask)	Δ in that subgroup
OP1 max. balanced accuracy	0.781 $\rightarrow$ 0.408	-0.373 [-0.508, -0.243]	+0.239 [+0.155, +0.345]	0.779 $\rightarrow$ 0.269	-0.510 [-0.660, -0.346]
OP2 specificity $\geq$ 0.80	0.747 $\rightarrow$ 0.371	-0.376 [-0.462, -0.297]	+0.225 [+0.163, +0.291]	0.744 $\rightarrow$ 0.223	-0.521 [-0.642, -0.382]
OP3 specificity $\geq$ 0.90	0.542 $\rightarrow$ 0.218	-0.324 [-0.419, -0.228]	+0.143 [+0.098, +0.198]	0.526 $\rightarrow$ 0.108	-0.418 [-0.536, -0.276]
OP4 sensitivity $\geq$ 0.80	0.783 $\rightarrow$ 0.423	-0.360 [-0.439, -0.257]	+0.232 [+0.159, +0.302]	0.779 $\rightarrow$ 0.267	-0.513 [-0.627, -0.371]
OP5 sensitivity $\geq$ 0.90	0.888 $\rightarrow$ 0.646	-0.242 [-0.345, -0.162]	+0.240 [+0.159, +0.345]	0.867 $\rightarrow$ 0.510	-0.356 [-0.497, -0.225]

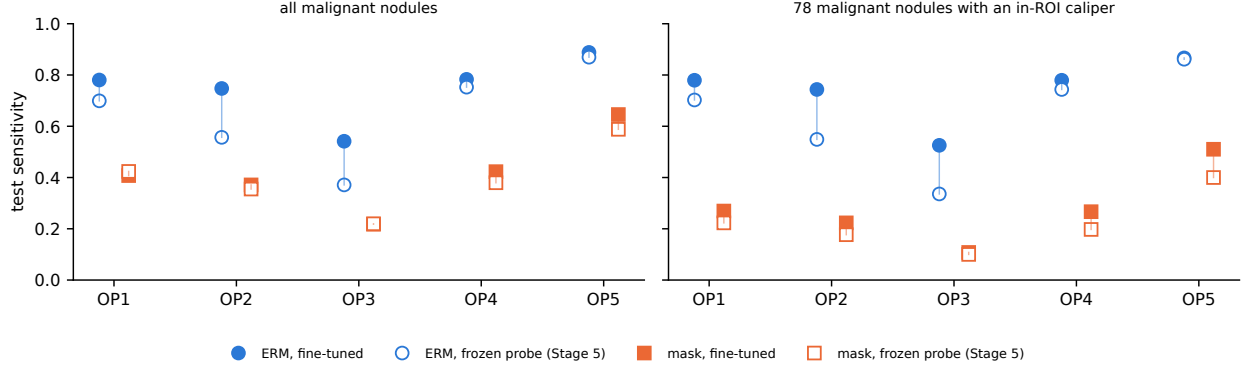


Figure 2: Test sensitivity of ERM (circles) and of the masked model (squares) at the five validation-fixed operating points, for the fine-tuned model (filled) and the frozen probe of Stage 5 (open); left: all malignant nodules, right: those with a caliper on the nodule. Thin lines join the two models.

5.7 Round 7: is the loss a threshold artefact?

Table 10. With each model’s threshold re-tuned on the test set so that both reach a test specificity of 0.80, the overall sensitivity of the two arms is the same (fine-tuned -0.017 $[-0.124, +0.070]$), but the conflicting malignant nodules still lose: fine-tuned $0.544 \rightarrow 0.390$ (-0.154 $[-0.302, -0.023]$; TM1 supported, Holm $p = 0.019$), frozen probe $0.505 \rightarrow 0.331$ (-0.174 $[-0.313, -0.012]$; TM2 supported). The aligned malignant nodules tend to gain (fine-tuned $+0.051$ $[-0.070, +0.150]$, frozen probe $+0.124$ $[+0.001, +0.231]$): masking moves detections from the nodules whose caliper points to benign to those whose caliper points to malignant, as the location law predicts. The surviving loss is about 30% of the loss at validation-fixed thresholds (FT3), so most of that larger loss came from transporting thresholds; the part that remains cannot be removed by any threshold. At specificity 0.90 and at matched sensitivity the direction is the same with intervals that include zero. On the other natural test sets (Table 11, descriptive) the conflicting positives lose in BCN (-0.040 $[-0.065, -0.015]$), gain in HAM ($+0.067$ $[+0.032, +0.106]$), matching the hard-pair AUROC of Stage 5, and show no change in MSK and capsule.

Table 10: Round 7: sensitivity when each arm’s threshold is chosen on the test set itself, so that both arms reach the same test specificity (or sensitivity); ERM $\rightarrow$ mask, and mask $-$ ERM with crossed 95% CIs. The first row of each model is the registered test (TM1, TM2).

Model	Both arms matched at	Malignant, in-ROI caliper (78)	Malignant, no in-ROI caliper	All malignant	Specificity ERM / mask
fine-tuned	test specificity 0.80	0.544 $\rightarrow$ 0.390, -0.154 $[-0.302, -0.023]$	0.563 $\rightarrow$ 0.614, $+0.051$ $[-0.070, +0.150]$	0.557 $\rightarrow$ 0.540, -0.017 $[-0.124, +0.070]$	0.806 / 0.804
	test specificity 0.90	0.323 $\rightarrow$ 0.218, -0.105 $[-0.221, +0.019]$	0.343 $\rightarrow$ 0.437, $+0.094$ $[-0.028, +0.196]$	0.336 $\rightarrow$ 0.364, $+0.028$ $[-0.070, +0.119]$	0.903 / 0.904
	test sensitivity 0.80	0.803 $\rightarrow$ 0.715, -0.087 $[-0.172, +0.018]$	0.800 $\rightarrow$ 0.843, $+0.043$ $[-0.009, +0.083]$	0.801 $\rightarrow$ 0.801, $+0.000$ $[-0.005, +0.006]$	0.611 / 0.542
frozen probe	test specificity 0.80	0.505 $\rightarrow$ 0.331, -0.174 $[-0.313, -0.012]$	0.514 $\rightarrow$ 0.638, $+0.124$ $[+0.001, +0.231]$	0.511 $\rightarrow$ 0.536, $+0.025$ $[-0.083, +0.132]$	0.803 / 0.803
	test specificity 0.90	0.231 $\rightarrow$ 0.151, -0.079 $[-0.244, +0.086]$	0.313 $\rightarrow$ 0.390, $+0.077$ $[-0.040, +0.200]$	0.286 $\rightarrow$ 0.311, $+0.025$ $[-0.085, +0.141]$	0.903 / 0.904
	test sensitivity 0.80	0.779 $\rightarrow$ 0.713, -0.067 $[-0.155, +0.026]$	0.811 $\rightarrow$ 0.844, $+0.033$ $[-0.013, +0.075]$	0.801 $\rightarrow$ 0.801, $+0.000$ $[-0.006, +0.006]$	0.538 / 0.515

Table 11: Round 7, descriptive: the other natural test sets of Stage 5 at matched test specificity 0.80, mask – ERM (crossed 95% CIs); conflicting = positives whose artifact status conflicts with the test set’s own association.

Test set (frozen probe)	Conflicting positives	Aligned positives	All positives
Dermoscopy, BCN held out	−0.040 [−0.065, −0.015]	−0.034 [−0.062, −0.003]	−0.038 [−0.057, −0.017]
Dermoscopy, HAM held out	+0.067 [+0.032, +0.106]	+0.039 [−0.019, +0.095]	+0.059 [+0.027, +0.093]
Dermoscopy, MSK held out	−0.001 [−0.054, +0.053]	+0.193 [+0.067, +0.311]	+0.020 [−0.030, +0.070]
Capsule (held-out frames)	+0.037 [−0.013, +0.110]	+0.144 [+0.107, +0.196]	+0.131 [+0.098, +0.179]

Against the frozen probe Table 12. Every effect is at least as large for the stronger model: the hard-pair harm (−0.095 [−0.165, −0.022] → −0.102 [−0.178, −0.026]), the OP2 sensitivity loss, and the subgroup loss (−0.372 [−0.499, −0.236] → −0.521 [−0.642, −0.382]). The trap crossover is smaller than the frozen probe’s (+0.239 [+0.196, +0.280]) and larger than that of the fine-tuned ViT-S of Stage 5 (+0.122 [+0.039, +0.211]).

Table 12: The same analyses for the frozen probe of Stage 5 and for the fine-tuned model of this stage. The subgroup was post hoc for the probe and registered in advance for the fine-tuned model.

Quantity	Frozen DINOv2 ViT-B/14 probe (Stage 5)	Fine-tuned ConvNeXt-T (IN-22k) (this stage)
ERM test AUROC	0.735	0.769
Hard pairs, mask – ERM	−0.095 [−0.165, −0.022]	−0.102 [−0.178, −0.026]
OP2 sensitivity, mask – ERM	−0.202 [−0.300, −0.100]	−0.376 [−0.462, −0.297]
OP2 sensitivity, 78 malignant with in-ROI caliper	0.549 → 0.177	0.744 → 0.223
change	−0.372 [−0.499, −0.236]	−0.521 [−0.642, −0.382]
Thyroid trap crossover	+0.239 [+0.196, +0.280]	+0.187 [+0.111, +0.268]

6 What failed or changed

- **Hair presence.** The registered hair-free threshold (≤ 30 hair pixels) is far more sensitive than DermArtifactDB’s image-level label ($\kappa = 0.076$ [0.070, 0.082]), about half of the trap images have no hair by DermArtifactDB, and the disagreement differs by diagnosis. The hair position is right where hair exists.
- **Thyroid HA2 not testable.** The registered “any source” rule removed 87% of Trap B, because the two caliper labellers err in opposite directions; the consensus analysis (E1) is exploratory.
- **FT0** is missed by 0.004 AUROC: the fine-tuned model is near, not at, the benchmark.
- **Capsule** labels could not be checked (29 frames); **ovary** labels wait for the rating, because BUSClean is invalid on ovarian screens. Their “supported” HA2 rows reproduce Stage 3 and are not new evidence.
- **The rating** (A4, A4b) has not been done.
- **Threshold transport.** About 30% of the subgroup loss at validation-fixed thresholds survives thresholds re-tuned on the test set (round 7); the rest was a consequence of transporting thresholds. The secondary matched points (specificity 0.90, sensitivity 0.80) point the same way without significance.
- **Reporting corrections after the run** (no estimate changed): the attenuation correction had been printed per source even where another source of the same cohort flagged differential error, which A3 does not allow; it is now withheld for both flagged cohorts (dermoscopy, thyroid). For thyroid the withdrawn MedGemma-based value would have been more than 2.5 times the observed crossover, because $1 - e_A - e_B$ is small when a noisy labeller is taken as the truth. e_A and e_B against DermArtifactDB, which has no location information, had been printed as 0 and

are now not defined. κ with the Kabir masks is not informative (our masks mark hair on all 485 images) and is replaced by the raw agreement. A committed smoke-test output was removed.

7 Anticipated questions

If DermArtifactDB finds no hair on half of the trap images, are the Stage 3 hair results wrong? No. Those images carry weak or no hair, so they dilute the trap; removing every contradicted image makes the crossover larger ($+0.147 [+0.111, +0.185] \rightarrow +0.199 [+0.164, +0.233]$), and the position of the hair, which is what the law is about, is confirmed by two independent sources ($e_A + e_B \leq 0.025$). The controlled experiments of Stages 2 and 4 place the artifact themselves and do not use these labels at all.

How can κ be so low if the artifact-free cell is almost perfectly clean? Because the disagreement is one-sided. Thirty native pixels are a short stretch of a single hair, or a dark line of the lesion, while DermArtifactDB records whether hair is visible as an artifact. κ is dominated by the many images we call hair-bearing and DermArtifactDB calls hair-free.

Could the differential error have produced the crossover? It cannot be ruled out from the error rates alone, which is why the registered rule sends a flagged cohort to A2: without the contradicted images the crossover is larger, not smaller.

Why did the thyroid cleaning fail if both labellers are informative? “Informative” means $\kappa \geq 0.40$, not unbiased. MedGemma says “yes” too often and BUSClean too rarely; a rule that lets either one veto an image removes most images whenever the two disagree, and they disagree most on calipers outside the nodule.

Is MedGemma a credible labeller? It is a general medical vision–language model, not a caliper detector, and is used only through the informativeness rule. Its answers are fixed-prompt, greedy and local, it never sees our masks, and it is judged against other labels, not trusted: informative for thyroid, undecided for ovary, where its agreement with our detector is low.

Why is the attenuation correction not reported? The registration allows it only for cohorts without differential error, and both cohorts with an informative source are flagged. The correction also treats the source as the truth; with a noisy labeller it inflates the crossover rather than correcting it.

Is 0.769 “benchmark-level”? Not by the registered rule, which is 0.773. The paper can call the model near-benchmark (0.004 below the lowest published baseline, above the frozen probe) and must not call it benchmark-level.

Why does sensitivity fall by half while the AUROC barely moves? Mostly because of the threshold: the operating points transport a threshold from validation to test, and training on masked images moves the model’s test scores relative to it (specificity rises at every operating point). Round 7 separates the two parts: with thresholds re-tuned on the test set the overall sensitivity is the same for both arms, and what remains is a loss of $-0.154 [-0.302, -0.023]$ among the malignant nodules whose caliper, the retained shortcut, points to benign.

Would re-tuning the threshold at the hospital remove the harm? Only in part. Re-tuning on the test set itself is the best any site can do, and the conflicting nodules still lose sensitivity (TM1, TM2).

Why use exploratory analyses at all? Because the registered rule failed for a structural reason that the consensus view makes visible. E1 and E2 are reported apart from the registered results and are never tests.

8 Conclusion

Supported. Where an artifact is present, our labels put it on the correct side of the ROI boundary (dermoscopy, two independent sources); the dermoscopy crossover survives, and grows, on uncontradicted labels (HA2); a fine-tuned model near the benchmark shows every registered harm of masking — on conflicting pairs (FT1), at every operating point (FT2), in the subgroup of 78 malignant nodules with an in-ROI caliper, registered in advance (FT3), and in the traps (FT4); the subgroup loss survives thresholds re-tuned on the test set (TM1, TM2). **Qualified.** Hair presence is over-called, with errors that differ by diagnosis; thyroid caliper labels agree with two independent labellers where those agree with each other, weakest for calipers outside the nodule, but the registered cleaned-trap test could not run; the fine-tuned model is 0.004 below the benchmark; most of the operating-point loss is threshold transport, and the part that survives re-tuning (-0.154 [$-0.302, -0.023$]) has a wide interval. Exploratory: the thyroid crossover holds on consensus labels ($+0.257$ [$+0.209, +0.304$]) and the subgroup loss holds in the doubly confirmed nodules. **Not yet known.** Whether the ovary caliper labels are right, and the author’s blinded rating; the capsule labels cannot be checked with public data.

9 Questions for the supervisor

1. Is a clinician-free validation — public expert annotations, two independent automatic labellers and a blinded non-clinician rating — acceptable for the paper, stated as such?
2. Should the paper lead with the cleaned dermoscopy crossover and report the over-called hair presence as a limitation of the published hair masks at our threshold?
3. May the thyroid clinical result now be stated as registered (FT3, TM1), with the model described as near-benchmark and the operating-point loss split into its threshold part and its surviving part?
4. Is the author’s rating worth doing now, given that the ovary labels depend on it, or should the ovary real-artifact evidence be presented as unvalidated?

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